Exercise DL

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Objective

This exercise aims to develop a model (machine learning or deep learning) to predict the engagement level generated by different tourist points of interest (POIs).

Exploratory data analysis

Data contains 1,569 samples and available features could be splitted in three classes.

- **Engagement Features**: These columns contain information used to build the target feature that classifies POI engagement (**Likes**, **Dislikes**, **Bookmarks**, and **Visits**).
- **Metadata Features**: Columns that could be used as model inputs with simple transformations or encoding (**categories**, **tier**, **locationLon**, **locationLat**, **tags**, and **xps**).

- **Description Feature**: Text descriptions of POIs can be found in the **shortDescription** column. Text descriptions of the POIs are contained in the shortDescription column. As this feature cannot be processed alongside the metadata features, it will be analyzed separately.
- **Image Data**: The path to images associated with each POI (**main_image_path**).

The dataset is complete and free of duplicates.

Engagement features.

All these features are quantitative variables. As shown in Figure 1, features **Likes**, **Dislikes** and **Bookmarks** show a trimodal distribution. In all three distributions, two sharp peaks appear on the left side and a flat peak appears on the right side. **Visits** shows an unimodal distribution with positive skewness.

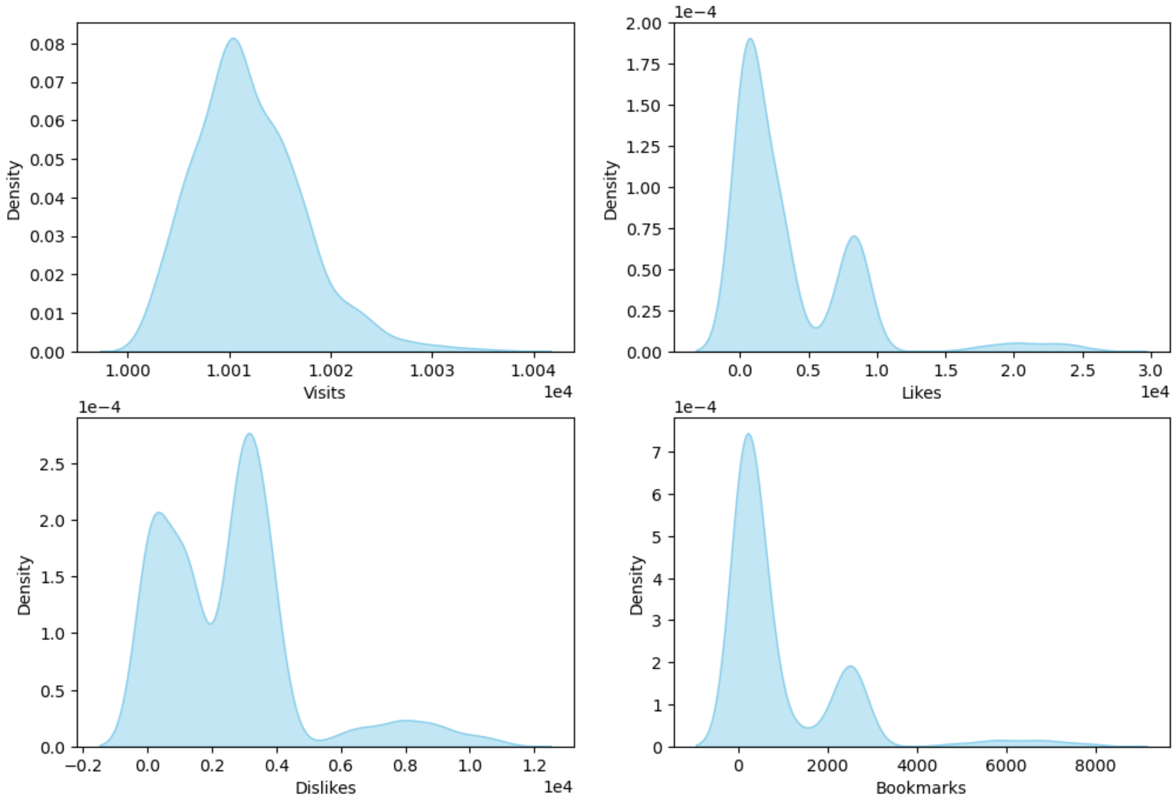

Figure 1: Density plots for each engagement feature

It is reasonable to create a new feature from the difference between **Likes** and **Dislikes**. Negative values would indicate low engagement and positive values would suggest high engagement.

This new feature show a multimodal distribution with several sharp peaks on the left and a flat peak on the right (see figure Figure 2)

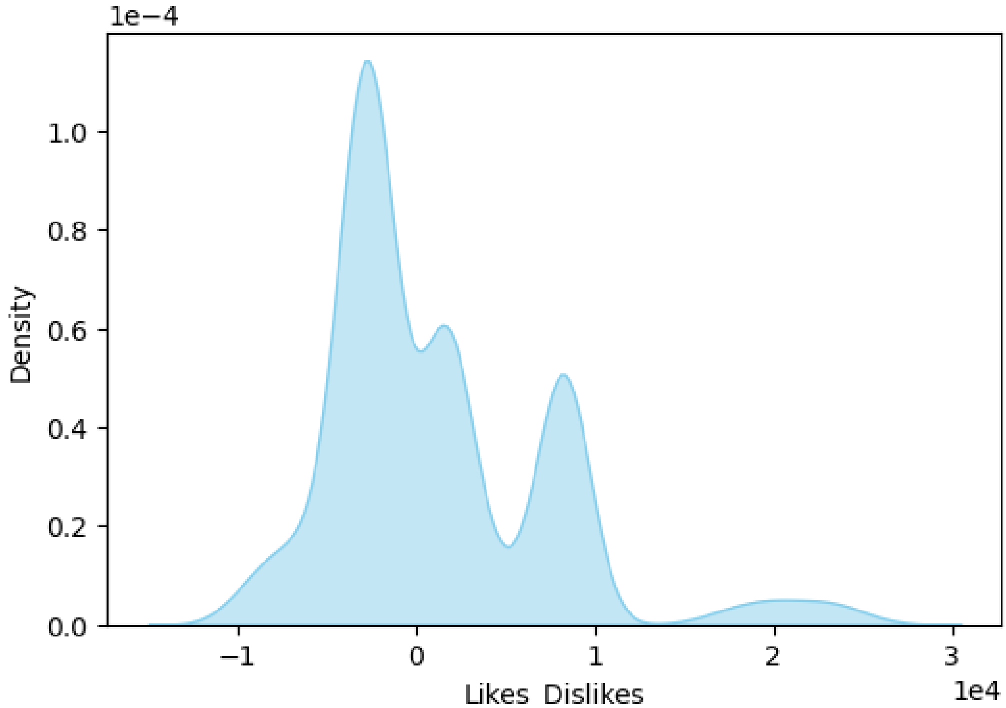

Figure 2: Density plots for new feature (Likes - Dislikes)

Correlation analysis shows strong relationships between the features **Likes**, **Dislikes**, **Likes_Dislikes**, and **Bookmarks** (Figure 3). **Visits** has low correlation with the other features.

Metadata features

The majority of samples (1,073) are assigned three labels in the **categories** feature, while only two POIs have none (Figure 4).

Across the dataset, 12 distinct labels are used. The most common are 'Historia' and 'Cultura,' each appearing in over half of all POIs (Figure 5). In contrast, labels such as 'Pintura,' 'Naturaleza', 'Cine' and 'Gastronomía' are relatively rare, each found in fewer than 5% of the records.

The **tags** feature provides information similar to categories, but in greater detail. The dataset contains 2,935 unique tags, most of which are applied to more than eight POIs (Figure 6).

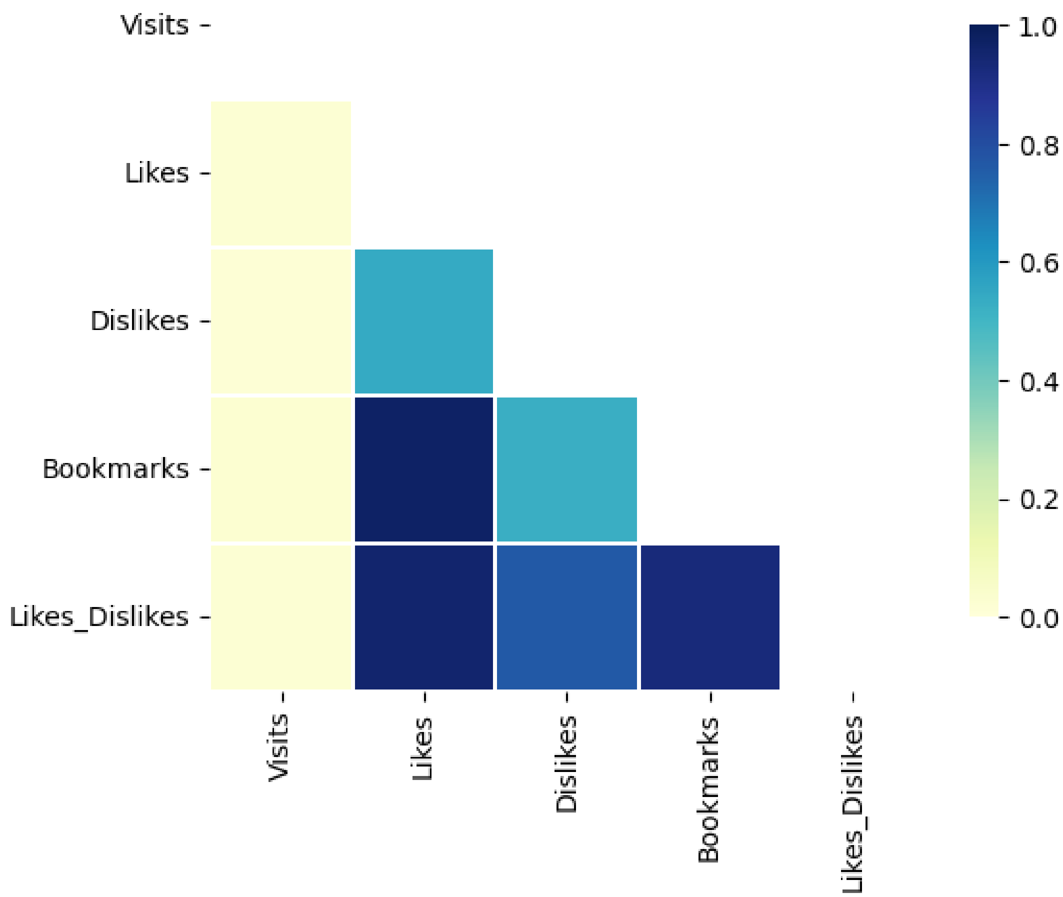

Figure 3: PCC correlation between engagement features

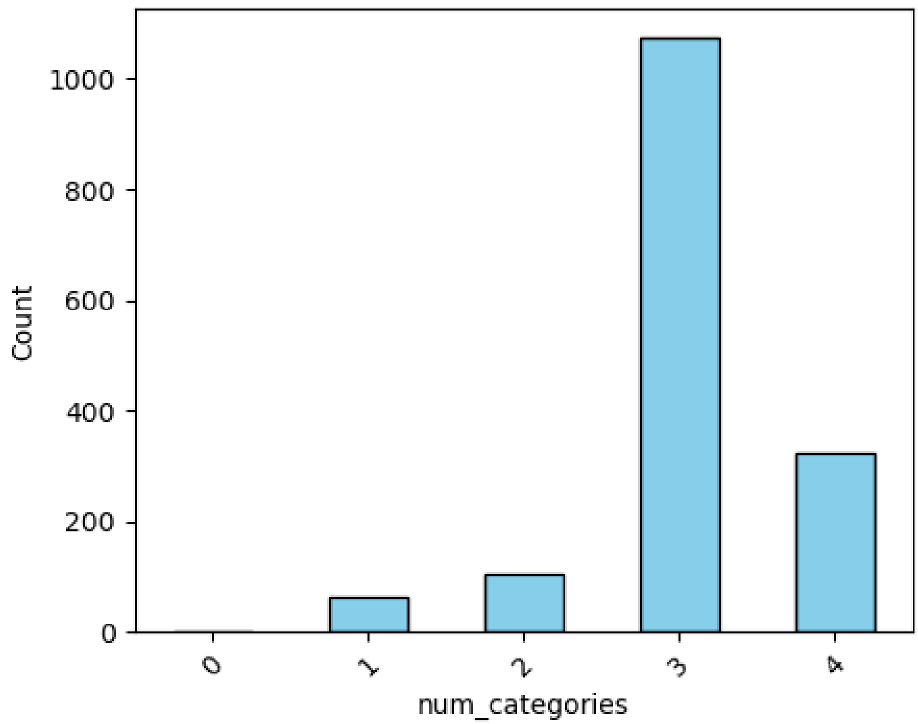

Figure 4: Distribution of number of categories

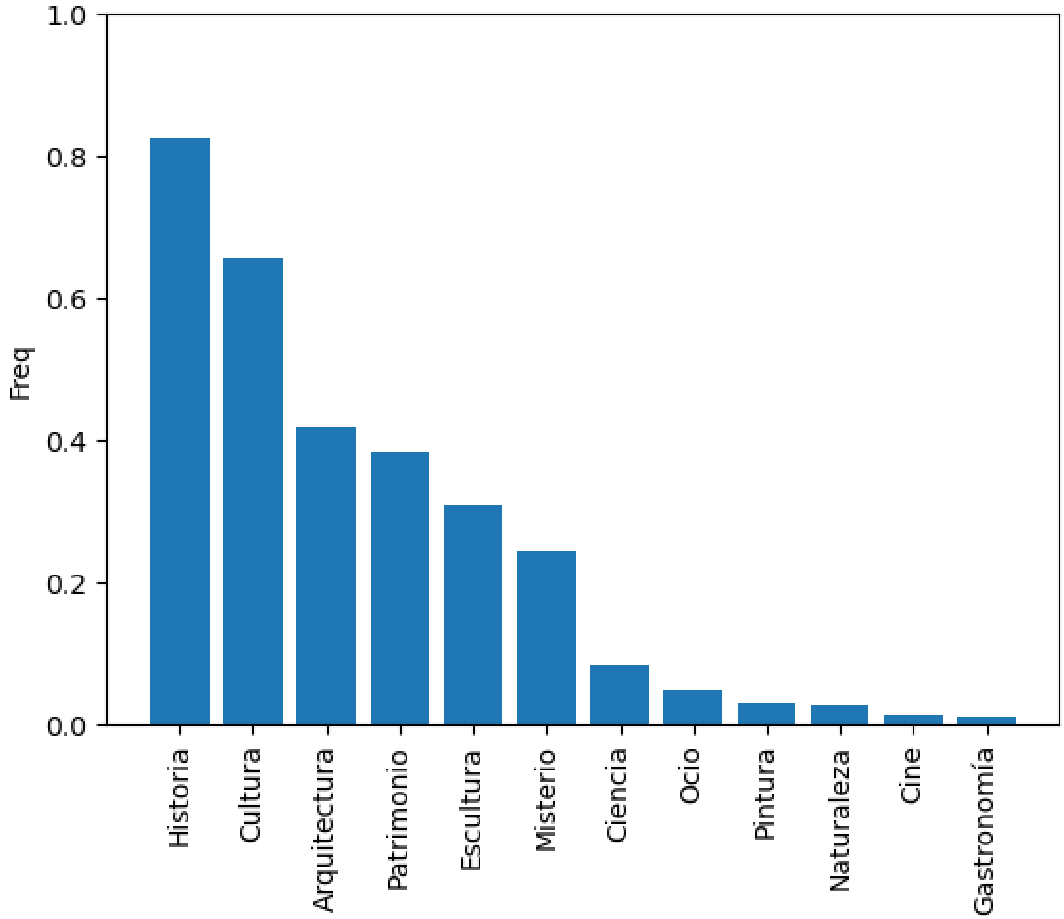

Figure 5: Frequencies for the categories

However, a significant number of tags are used infrequently, appearing in only a few POIs—sometimes just one.

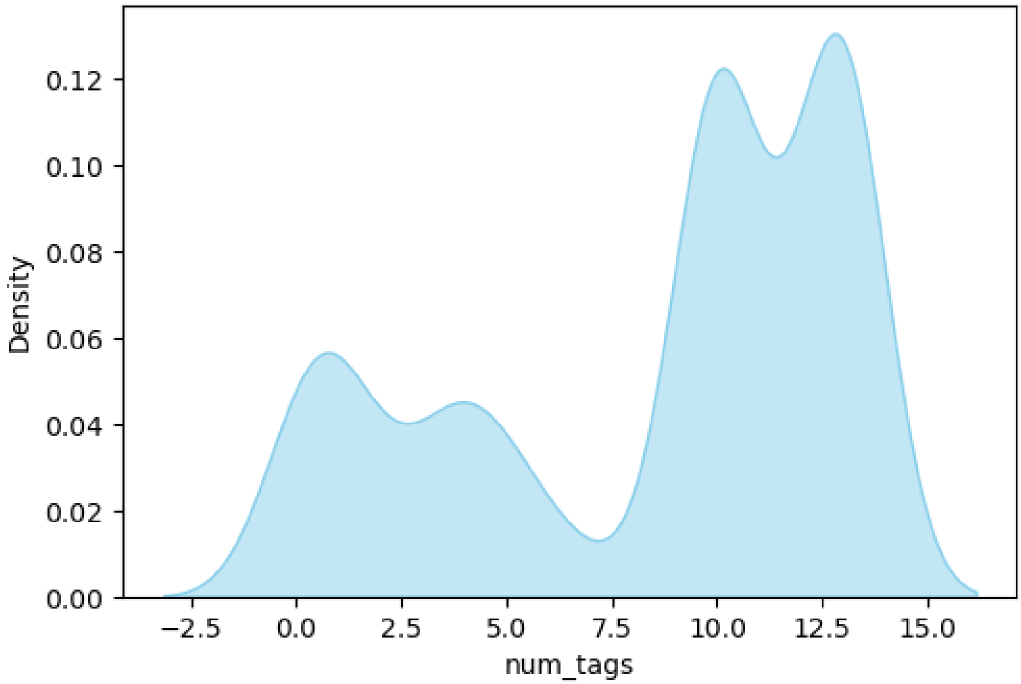

Figure 6: Distribution of number of tags

As shown in Figure 7, the majority of POIs have a **tier** value of one. The number of POIs decreases as the **tier** increases to four, with a pronounced drop occurring between tiers two and three.

The features locationLon and locationLat exhibit distributions with sharp peaks at 0 and 40, respectively (Figure 8). This pattern suggests that most POIs are concentrated within a small geographic region. Meanwhile, the xps feature displays a multimodal distribution characterized by a sharp peak centered at 1000.

Description feature.

A short text describing the POIs is available at **shortDescription** feature. It could be in the range from two words to a maximum of 89 words (Figure 9).

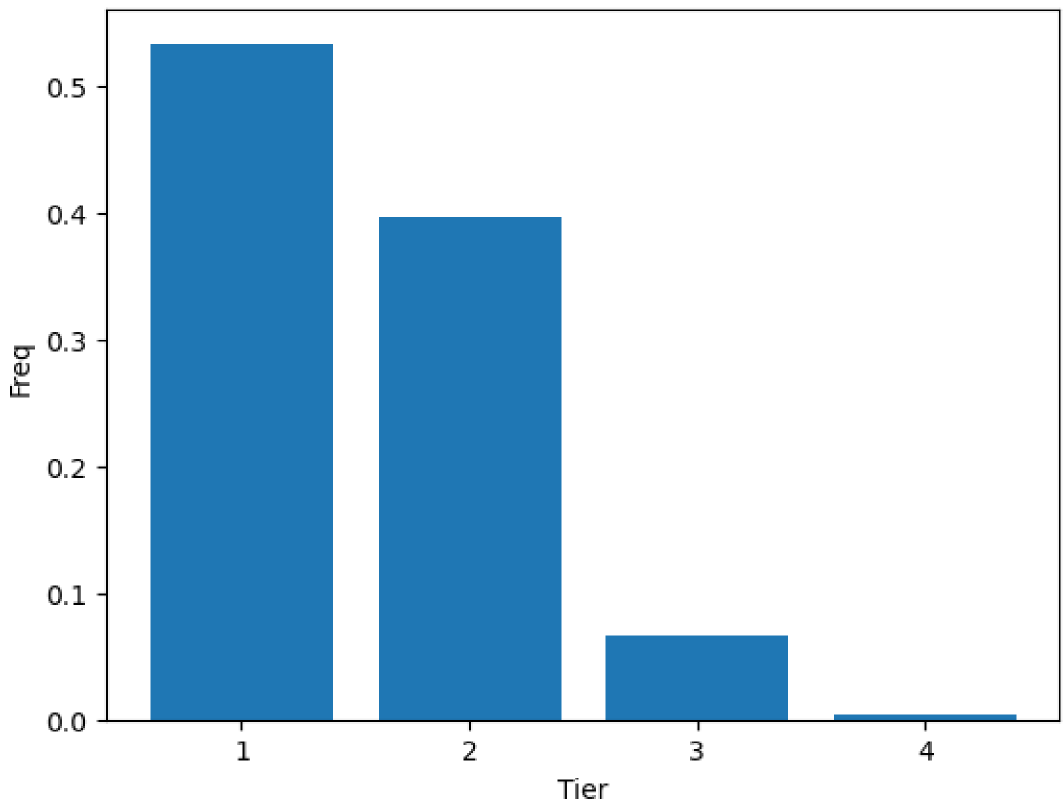

Figure 7: Frequencies for feature tier

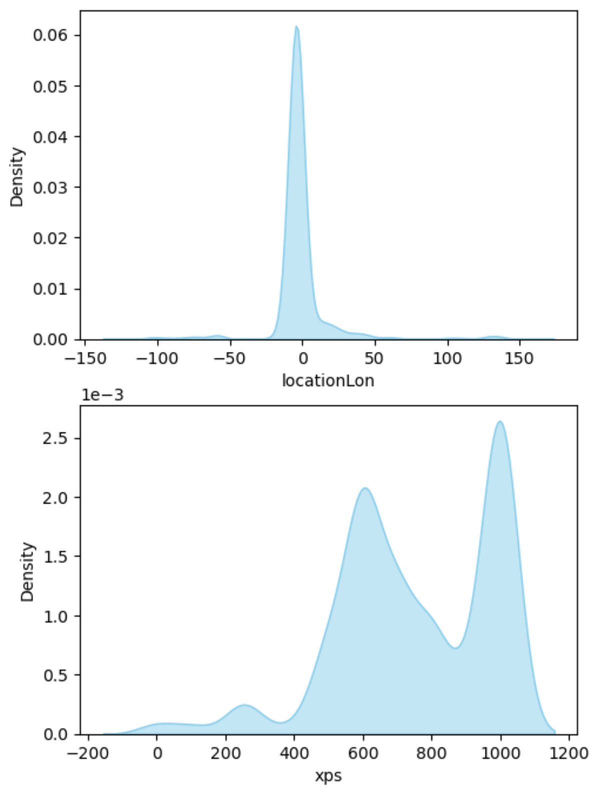

Figure 8: Distribution for features locationLon, locationLat and xps

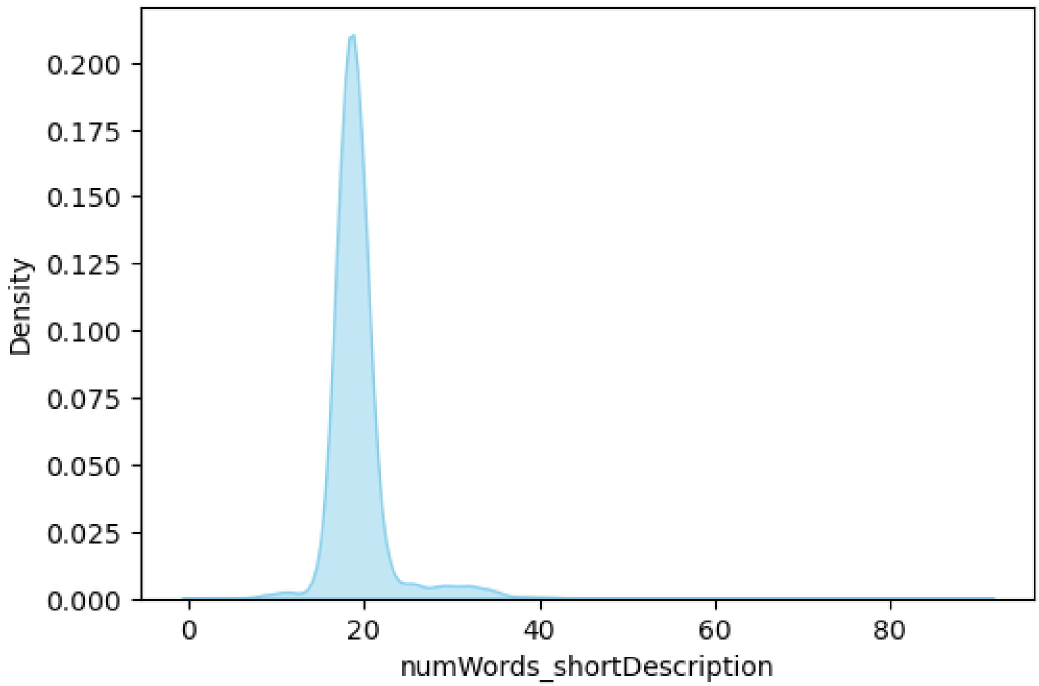

Figure 9: Distribution for number of words in shortDescription

Image feature.

Figure 10 displays a random sample of images. The distribution of the standard deviation (SD) of pixel values for each channel was analyzed and found to be approximately normal. However, some images exhibit very low SD values (Figure 11). Upon inspection, these were found to be 'empty' images, appearing almost entirely white (Figure 12).

Figure 10: Example of images of POIs

Target feature

Both the Bookmarks and Likes_Dislikes features are reliable indicators of user engagement. They exhibit a high correlation, so either would be a suitable choice for the analysis.

In this study, Likes_Dislikes was selected because it allows for a straightforward separation of samples into low and high engagement based on the sign of its value. Samples with negative values are classified as low engagement POIs, while those with positive values are classified as high engagement POIs. Using this threshold, 827 POIs were identified as low engagement and 742 as high engagement.

The Visits feature could be used to normalize Likes_Dislikes, as shown in Figure 13, to create an engagement rate metric.

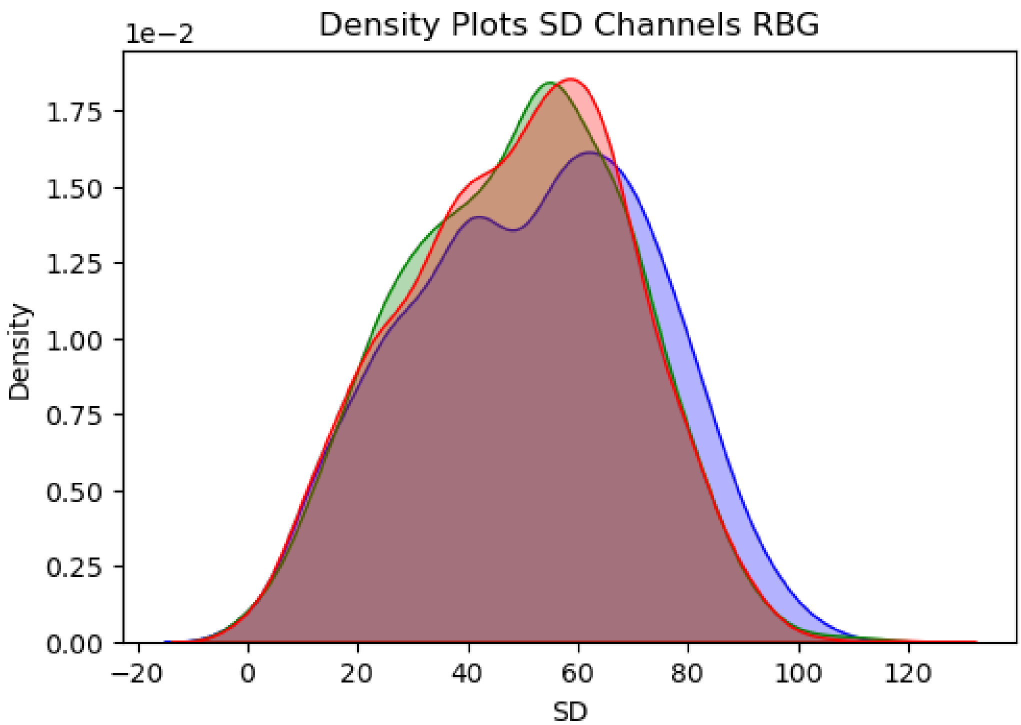

Figure 11: Distribution of SD for each channel of the images

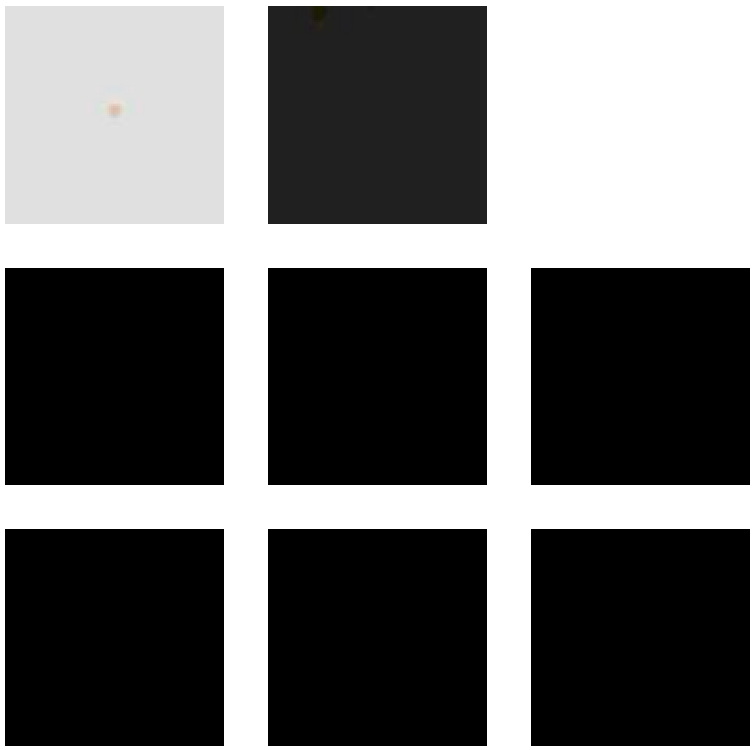

Figure 12: Example of images without any information

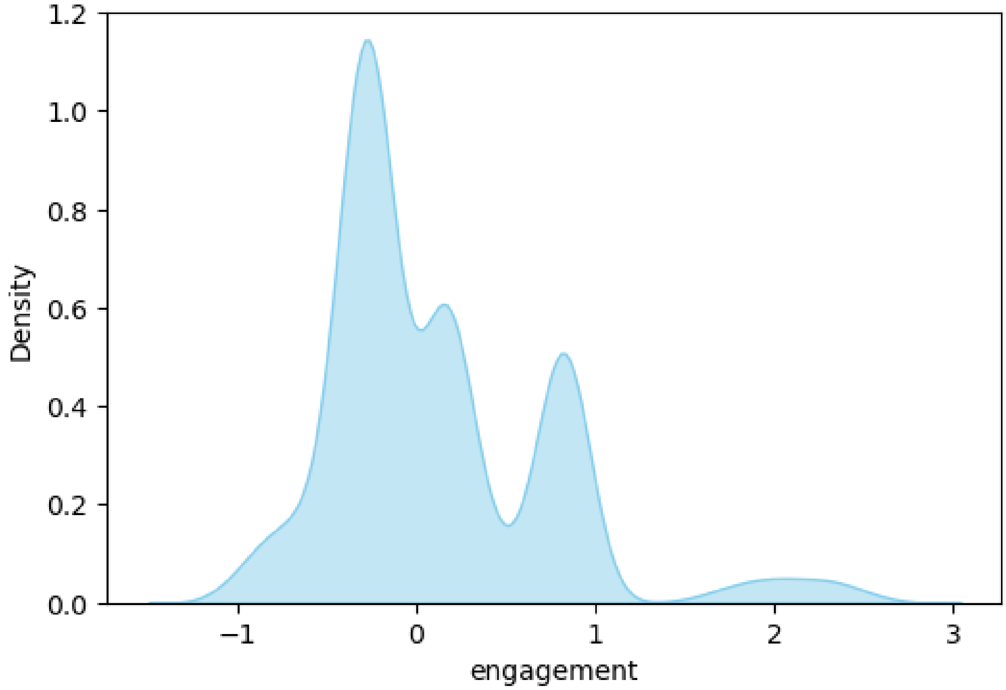

Figure 13: Density plots for each engagement feature

Optimization strategy

Hyperparameter tuning was performed using Optuna framework exploring each model's parameter space to identify optimal configurations that maximize predictive performance using TPE algorithm. Several scores were calculated but F1-score was used to drive hyperparameter optimization.

Machine learning models

Machine learning models trained on metadata features consistently achieved an F1-score of ~0.8. This performance was characterized by high sensitivity (>0.97) in identifying high-engagement POIs but low precision (<0.68), leading to a substantial number of false positives.

Model	Train score	Valid score	Precision	Sensitivity
Log Regression	0.805	0.796	0.675	0.971
Random Forest	0.807	0.807	0.677	1
XGBoost	0.809	0.803	0.675	0.992

Deep learning models

Multiple deep learning models were developed and assessed using different combinations of metadata, images, and embeddings.

FCNN Shallow: A shallow, fully connected neural network that uses metadata as input.

FCNN PCA: fully connected neural network with progressive compression architecture. It uses embeddings from the shortDescription field as input.

ResNet18 Pre: A standard ResNet18 model with pretrained weights, using images as input. All layers are frozen.

ResNet18 Layer4: A ResNet18 model with pretrained weights, fine-tuned for the task. The weights of layer4 are unfrozen and optimized, while earlier layers remain frozen. The model uses images as input.

Multi Modal Pre: Multi-modal neuronal network with three input branches.

- Image Branch: Processes images through a ResNet18 backbone.
- Text Branch: Processes shortDescription embeddings through a fully connected network with a progressive compression architecture.
- Metadata Branch: Directly incorporates metadata features.

The outputs from all three branches are concatenated and passed through a final classifier.

Multi Modal Layer4: Same as above model but with optimization of weights of layer 4.

Model	Train score	Valid score	Precision	Sensitivity
FCNN Shallow	0.748	0.79	0.669	0.966
FCNN PCA	0.309	0.643	0.474	1
ResNet18 Pre	0.714	0.8	0.726	0.891
ResNet18 Layer4	0.967	0.851	0.862	0.84
Multi Modal Pre	0.892	0.878	0.91	0.849
Multi Modal Layer4	0.874	0.849	0.849	0.849

FCNN Shallow and **ResNet18 Pre** yield results comparable to traditional Machine Learning models. Using the pre-trained ResNet18 via transfer learning achieves a balance between precision and sensitivity, improving precision at the cost of some sensitivity reduction. **FCNN_pca** demonstrates poor performance, inferior to all ML models.

ResNet18 Layer4, **Multi Modal Pre**, and **Multi Modal Layer4** models achieve significantly higher F1-scores, all exceeding 0.84. However, **ResNet18 Layer4** shows substantial overfitting, with a high training F1-score that doesn't generalize to validation. In contrast, both Multi Modal models show consistent performance across training and validation sets. **Multi Modal Pre** delivers the best overall results, with precision exceeding 0.9.

The choice between top-performing models depends on the specific objective. For higher precision in detecting high-engagement POIs, **Multi Modal Pre** is recommended. When higher sensitivity is required for this task, **Random Forest** becomes the more suitable choice.